



Department of Computer Science and Engineering

Academic year 2025-2026 (Odd Semester)

CIE III: Question Paper

Course	Operating Systems		
Date	JAN 2026	Maximum Marks	60
Course Code	CS235AI	Duration	120 Min
Sem	III	CIE – III	

SL No.	Part A	M	BT	CO
1.1	Formulate how long a paged memory reference takes if memory reference takes 200 nanoseconds. Assume a paging system with page table stored in memory.	01	2	1
1.2	Consider a logical address space of 64 pages of 1024 words each, mapped onto a physical memory of 32 frames. a. How many bits are there in the logical address? b. How many bits are there in the physical address?	02	2	2
1.3	What is compaction? Explain with an Example	02	1	1
1.4	In the working set model, for: 2 6 1 5 7 7 7 5 1 6 2 3 4 1 2 3 4 4 4 3 4 3 4 4 4 1 3 2 3, if Delta = 10, then the working set at time t1(...751) is what?	02	3	2
1.5	A file of size 80 KB is stored on a disk with block size 1 KB. How many blocks are required? An index block can store 256 pointers. Block size is 4 KB. What is the maximum file size?	02	3	3
1.6	How does OS utilize swapping technique?	01	1	2

	Part B			
2.a	Given a system that runs large database applications and modular software systems, analyse whether paging or segmentation is more suitable. Justify your choice by discussing fragmentation, protection mechanisms, and logical program structure.	06	2	3
2.b	A company is designing a memory management system for its new OS that will run: 1. A database server storing large, fixed-size records. 2. A modular software development environment with dynamically loaded/unloaded modules. The system designer is considering paging vs. segmentation : a) Identify which type of fragmentation (internal or external) is likely for each workload and explain why. b) Discuss how each fragmentation type would impact memory utilization and system performance, given the database grows continuously and modules are loaded/unloaded frequently.	04	2	4
3	A system has memory partitions of sizes: 100 KB, 500 KB, 200 KB, 300 KB, 600 KB. Four processes with sizes 212 KB, 417 KB, 112 KB, 426 KB need to be allocated memory. a) Allocate the processes to memory partitions using the following allocation strategies: First-Fit, Best-Fit, Worst-Fit b) Show the allocation step-by-step and indicate any processes that cannot be allocated. c) Analyze which allocation strategy is most efficient in terms of memory utilization and briefly justify your answer.	10	3	3

4.a	Discuss situation under which the most frequently used page replacement algorithm generates fewer page faults than the least frequently used page replacement algorithm. Also discuss under which circumstances the opposite holds.	06	2	2
4.b	A system uses copy-on-write for virtual memory. Analyze a scenario where multiple processes share a large read-only dataset. How does COW improve memory utilization in this case? When multiple processes share a large read-only dataset, copy-on-write (COW) allows them to use the same physical pages without duplication, marking them as read-only. A private copy is created only if a process modifies a page, minimizing memory usage while maintaining process isolation.	04	2	3
5.a	Discuss the role of various file attributes in operating systems, explaining how they contribute to security, system auditing, recovery, and efficient file operations. In your answer, include ownership, permissions, timestamps, file size, location, and extended attributes.	04	2	2
5.b	Compare sequential, direct, and indexed sequential file access methods in terms of efficiency and use cases. Clarify why sequential access remains widely used, why indexed sequential is preferred in databases, and the efficiency differences between linked and indexed allocation for random access	06	3	3
6	How does demand paging enable efficient memory usage in real-world systems such as multitasking operating systems, and what hardware support is required to make demand paging possible? A program has 10 pages but only 4 page frames are available. Explain how demand paging decides which pages to load into memory when a page fault occurs.	8+2	4	4

Course Outcomes

CO 1	Demonstrate the fundamental concepts of operating system like process management, file management, memory management and issues of synchronization.
CO 2	Analyze and interpret operating system concepts to acquire a detailed understanding of the course.
CO 3	Apply the operating systems concepts to address related new problems in computer science Domain.
CO 4	Design or develop solutions using modern tools to solve applicable problems in operating systems domain.
CO5	Extend the theoretical knowledge acquired through the course to demonstrate skills like investigation, effective communication, working in team/Individual, following ethical practices by implementing operating system concepts/ applications and engage in life long learning.

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	3	17	26	14	-	3	27	20	10	-	-